

## Task 6

## Triggers, views and Exceptions

## AIM:

To implement triggers, views and Exceptions in SQL

## PROCEDURE:

1. Create customer table
2. Insert the values
3. Perform trigger operation
4. Perform view operation
5. Perform Exception Handling

## 1. Implementing Triggers

```
CREATE TABLE Customer (
    CustomerID INT NUMBER PRIMARY KEY,
    Name VARCHAR(50),
    Age INT,
    Phone VARCHAR(15),
    Email VARCHAR(50)
);
```

```
CREATE OR REPLACE TRIGGER prevent_underage_customers
BEFORE INSERT ON Customer
FOR EACH ROW
BEGIN
```

```
    IF :NEW.Age < 18 THEN
```

```
        RAISE_APPLICATION_ERROR(-20001, 'Customer age must be
        18 or above');
```

```
    END IF;
```

```
END;
```

2. Create a Log Table for customer Actions

```
CREATE TABLE CustomerLog (  
    LogID NUMBER PRIMARY KEY,  
    CustomerID INT,  
    ActionType VARCHAR(20),  
    ActionDate TIMESTAMP DEFAULT SYSTIMESTAMP  
);
```

2. Creating Views

View for customer Contact Information

```
CREATE OR REPLACE VIEW view_customerContacts AS  
SELECT CustomerID, Name, Phone, Email  
FROM Customer;
```

CREATE OR REPLACE VIEW view\_customerAgeCategory AS

SELECT

Customer ID,

Name,

CASE

WHEN Age BETWEEN 18 and 25 THEN 'Young'

WHEN Age BETWEEN 26 and 45 THEN 'Adult'

WHEN Age BETWEEN 46 and 65 THEN 'Middle Age'

ELSE 'Senior'

END AS AgeCategory

FROM Customer;



### 3. Exception Handling

Sorted Procedure with Exception Handling for Inserting Customer Records

```
CREATE OR REPLACE PROCEDURE InsertCustomer (
```

```
    p_customerID IN INT,
```

```
    p_Name IN VARCHAR,
```

```
    p_Age IN INT,
```

```
    p_Phone IN VARCHAR
```

```
    p_Email IN VARCHAR
```

```
)
```

```
IS
```

```
BEGIN
```

```
    INSERT INTO customer (customer ID, Name, Age, Phone, Email)
```

```
    VALUES (p_customerID, p_name, p_Age, p_phone, p_Email);
```

```
    DBMS_OUTPUT.PUT_LINE('Customer Record Inserted successfully');
```

```
EXCEPTION
```

```
    WHEN OTHERS THEN
```

```
        DBMS_OUTPUT.PUT_LINE('ERROR: ' || SQLERRM);
```

```
END;
```

~~Function~~ to Fetch Customer Details with Error Handling

```
CREATE OR REPLACE FUNCTION GetCustomerDetails (p_customerID IN NUM
```

```
RETURN VARCHAR
```

```
IS
```

```
    customer_info VARCHAR(200);
```

```
BEGIN
```

```
    SELECT 'Name: ' || Name || ', Age: ' || Age || ', Phone: ' || Phone ||
```

```
           ', Email: ' || Email
```

```
    INTO customer_info
```

```
    FROM customer
```

```
    WHERE customerID = p_customerID;
```

```
    RETURN customer_info;
```

EXCEPTION

WHEN NO\_DATA\_FOUND THEN

RETURN 'customer Not Found';

WHEN OTHERS THEN

RETURN 'Error:'

END;

| VELTECH                 |    |
|-------------------------|----|
| EX No.                  |    |
| PERFORMANCE (5)         | 6  |
| RESULT AND ANALYSIS (5) | 5  |
| VIVA VOCE (5)           | 5  |
| RECORD (5)              | 5  |
| TOTAL (20)              | 11 |
| DATE                    | 15 |
| WITH DATE               | 16 |

11/9/15

RESULT: Thus, Implementation of Triggers, views and exceptions operations are executed and completed successfully.